Free Hosting Plan for the Chess Platform

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You can run the whole platform for free on one Oracle Cloud Always Free server, but it will serve a small community, not a million players. This plan shows what runs where, how to set it up and what can go wrong.

1. What free hosting can and cannot do

Goal	Free?	Why
Build and test the platform	Yes	One free server is enough for development and load tests
Launch for friends or a small community	Yes, with care	My estimate is hundreds to a few thousand players at the same time. I have not tested this
1 million players at once	No	You would need many servers, and servers cost money

2. Where each part runs

Everything runs on one Arm virtual machine (VM) at Oracle. Oracle's free allowance for Arm machines is 2 CPUs and 12 GB of memory in total, plus 200 GB of disk.

Part	Where it runs	Why
Gateway, Game Service, Persistence Writer	The Arm VM (2 CPUs, 12 GB)	They fit the free allowance and talk to each other inside one machine, which is fast
Redis	The same VM	The free hosted Redis from Upstash gives only 256 MB and 500K commands a month. That is about 0.2 commands per second on average, which is too small

Part	Where it runs	Why
PostgreSQL	The same VM	The disk is 200 GB. The free hosted option from Neon gives 0.5 GB per project, and its compute stops when the monthly limit is reached
Event broker	The same VM, using NATS	NATS is open source, so there is no fee, and it is usually lighter than Kafka
Web client, if you have one	The same VM, or any free static host	It is only files
Backups	Oracle Object Storage	20 GB is free, and it keeps copies off the VM

3. Free offers and their limits

These limits were checked on the date at the top. Free offers change, so check them again before you launch.

Offer	What you get for free	Watch out for
Oracle Cloud Always Free	One Arm VM with 2 CPUs and 12 GB in total, two small AMD VMs with 1 GB each, 200 GB of disk, 20 GB of object storage, 10 TB of outbound data a month	Idle machines can be reclaimed. "Out of capacity" errors happen. Always Free resources must be in your home region
Google Cloud free tier	One e2-micro VM per month	It is small. A third-party guide says only some US regions qualify, so check Google's page
Neon	Postgres with 100 compute hours and 0.5 GB of storage per project	Compute stops at the monthly limit, and the first query after a pause is slow
Upstash	Redis with 256 MB and 500K commands a month	The command limit is very low

4. Setup steps

1. Create an Oracle Cloud account and pick your home region. Always Free resources must be created there.
2. Create one Arm VM (`VM.Standard.A1.Flex`) with 2 CPUs and 12 GB of memory, and use Ubuntu. If you get an "out of host capacity" error, try another availability domain or try again later.
3. Open the ports in two places: the network security list and the VM firewall. You need UDP 443 for QUIC, TCP 443 for the WebSocket backup and TCP 22 for SSH.
4. Build the Rust services for Arm (`aarch64`) and copy them to the VM.
5. Install Redis, PostgreSQL and NATS on the same VM. Cap Redis memory (for example at 1 GB) so it cannot take memory from the game workers.
6. Get a free TLS certificate (for example from Let's Encrypt) for your domain name. Browsers need a valid certificate to connect.
7. Run each service with `systemd` so it restarts after a crash or a reboot.
8. Back up PostgreSQL every day with `pg_dump` to Object Storage, and test that you can restore it.
9. Watch the machine with Oracle's free monitoring and with the metrics from the Monitoring section of the architecture document.

5. How many players it can hold

Moves are tiny and cheap to check, so the limits are the 2 CPUs, the memory and the number of open connections. My estimate is hundreds to a few thousand players at the same time. This is a guess until you load-test it.

A starting split of the 12 GB of memory (my estimate, not tested):

- **PostgreSQL:** about 3 GB.
- **Redis:** 1 GB.
- **NATS:** under 1 GB.
- **Gateway, Game Service and Persistence Writer:** about 4 GB.
- **Operating system and spare room:** about 3 GB.

Start with 1 or 2 game shards and leave the other CPU time to the Gateway, because encrypted QUIC connections use CPU. Watch the shard inbox size. If it keeps growing, you have reached the limit of this machine. Data transfer will not be the limit, because moves are only a few bytes and the free allowance is 10 TB a month.

6. Risks and how to handle them

Risk	What to do
Oracle can reclaim a machine it sees as idle. A machine counts as idle when CPU, network and (for Arm) memory use stay low for 7 days	A lightly used game server could look idle, so check its usage often. Keep backups and a quick way to rebuild
The free allowance can shrink. In 2026 Oracle cut the Arm allowance from 4 CPUs and 24 GB to 2 CPUs and 12 GB	Never size the system for more than 2 CPUs and 12 GB. Check the limits again before launch
One server is one point of failure	If it goes down, games stop. Rebuild from the daily backup. Free hosting has no automatic failover
"Out of capacity" when creating the VM	Try another availability domain, or try again later
A domain name usually costs money	Free subdomain services exist. You need a name to get the certificate
Neon pauses its compute after 5 minutes without use (source)	Use PostgreSQL on the VM for the Persistence Writer, and use Neon only if you want an off-site copy

7. When you outgrow it

You have outgrown the free server when the shard inbox keeps growing, memory stays near 12 GB or the CPUs stay busy. Take these steps in order:

1. Upgrade the Oracle account to Pay As You Go. Oracle says it only charges for use above the Always Free limits.
2. Move PostgreSQL to its own server.
3. Add more game servers and pick the owner of each game with consistent hashing (section 9 of the architecture document).

A million players at once needs many servers and a paid budget.

8. Sources

Checked on the date at the top.

- Oracle Cloud: Always Free Resources

- InfoQ: Oracle halves the Always Free Ampere A1 limits
- Neon pricing
- Prisma: Prisma Postgres vs Neon pricing 2026
- Upstash Redis pricing
- Google Cloud free tier
- Wikidocs: Google Cloud Always Free guide (third-party)